# Case Study: CAR0001

Model Analysis using Gemini 3.5 Flash

# CAR0001 — Speech-to-Text vs Real Transcript

Comparison of Ground Truth dialogue vs Gemini 3.5 Flash STT output (first ~100 words)

## 📄 Real Transcript (Ground Truth)

D: What brought you in today?
P: Sure, I'm I'm just having a lot of chest pain and and so I thought I should get it checked out.
D: OK, before we start, could you remind me of your gender and age?
P: Sure 39, I'm a male.
D: OK, and so when did this chest pain start?
P: It started last night, but it's becoming sharper.
D: OK, and where is this pain located?
P: It's located on the left side of my chest.
...

## 🎙️ Speech-to-Text Transcript (Gemini 3.5 Flash)

D: brought you in today? Sure I'm just having a lot of chest pain and so I thought I should get it
D: checked out. Okay and before we start could you remind me of your gender and age? Sure I'm 39
P: I'm a male. Okay and so when did this chest pain start? It started last night but it's becoming
D: sharper. Okay and where is this pain located? It's located on the left side of my chest. Okay and
...

# CAR0001 — Transcript Summary Comparison

Clinical summary extracted from Ground Truth vs Gemini 3.5 Flash STT output (first ~100 words)

## 📄 Ground Truth Summary

The patient, a 39-year-old male, presents with 8 hours of sharp, constant left-sided chest pain that worsens with lying down and deep breaths. He reports associated symptoms of lightheadedness, dyspnea, and mild palpitations, and noted visible neck swelling without accompanying pain.

## 🎙️ Gemini 3.5 Flash Summary (STT)

A 39-year-old male presents with sharp, left-sided chest pain of 8 hours duration, rated 7-8/10, which worsens when lying down and upon deep inspiration. The pain is accompanied by light-headedness, difficulty breathing, sweatiness, and mild palpitations. Given his significant family history of premature coronary artery disease (father had a myocardial infarction at age 45) and his heavy smoking history, an urgent cardiovascular and pulmonary workup is indicated.

# CAR0001 — Structured SOAP Report Comparison

| SOAP Field | Ground Truth | Predicted (Gemini 3.5 Flash) |
|---|---|---|
| **Chief Complaint** | Chest pain | Left-sided chest pain |
| **Symptoms** | sharp left-sided chest pain \| worsening pain on lying down \| worsening pain on deep breath \| lightheadedness \| dyspnea \| palpitations \| neck... | Left-sided chest pain \| Sharp pain \| Light-headedness \| Dyspnea (trouble breathing) \| Palpitations (racing of the heart) \| Sweating (seconda... |
| **Symptom Onset** | Last night (approximately 8 hours ago) | Started last night, lasting approximately 8 hours |
| **Medical History** | — | — |
| **Medications** | — | — |
| **Allergies** | none | — |
| **Exam Findings** | reported visible neck swelling \| patient reports 7-8/10 pain severity | Patient reports that his neck seems a little swollen \| Patient reports noisy breathing when experiencing severe difficulty breathing |
| **Diagnosis** | — | Chest pain of undetermined etiology |
| **Diff. Diagnosis** | Myocardial infarction (due to family history and symptoms) \| Pericarditis (suggested by positional component) \| Pneumothorax \| Musculoskelet... | Myocardial infarction \| Pericarditis \| Pleurisy \| Pulmonary embolism \| Musculoskeletal chest strain |
| **Prescribed Meds** | — | — |
| **Treatment Plan** | — | — |
| **Investigations** | — | — |
| **Follow Up** | — | — |

# Case Study: MSK0005

Model Analysis using Gemini 3.5 Flash

# MSK0005 — Speech-to-Text vs Real Transcript

Comparison of Ground Truth dialogue vs Gemini 3.5 Flash STT output (first ~100 words)

## 📄 Real Transcript (Ground Truth)

D: What brings you in?
P: Hi, I've I've had this pain on the outside of my uh right elbow now it's it I first started knowing, noticing it several months ago, but recently it's just been more painful.
D: OK, so you said several months ago. Um did anything happen several months ago? Was there any sort of trigger, trauma, anything like that to that area?
...

## 🎙️ Speech-to-Text Transcript (Gemini 3.5 Flash)

P: Hi, I've had this pain on the outside of my right elbow now. I first started noticing
P: it several months ago, but recently it's just been more painful.
D: Okay, so you said several months ago. Did anything happen several months ago? Was there
D: any sort of trigger, trauma, anything like that to that area?
P: No, there wasn't any trauma or any triggers that I noticed. I was just feeling it a bit
P: more at the end of work. Yeah, I was just feeling the pain a bit more at the end of
P: work.
...

# MSK0005 — Transcript Summary Comparison

Clinical summary extracted from Ground Truth vs Gemini 3.5 Flash STT output (first ~100 words)

## 📄 Ground Truth Summary

The patient is a healthy adult presenting with persistent right lateral elbow pain for several months, worsening with repetitive wrist movements at work. Examination findings are consistent with lateral epicondylitis, evidenced by pain on palpation of the lateral epicondyle and pain with resisted wrist extension and flexion. The patient was advised on activity modification, use of compression sleeves, continued over-the-counter analgesics, and physical therapy for recovery.

## 🎙️ Gemini 3.5 Flash Summary (STT)

A right-handed female patient presented with a several-month history of worsening lateral right elbow pain (rated 4/10), which is aggravated by repetitive wrist movements and holding heavy objects at her assembly line job. Guided self-palpation and physical assessment revealed tenderness localized to the lateral epicondyle and pain on wrist extension and flexion with the elbow extended. The findings are strongly indicative of lateral epicondylitis (tennis elbow), which will be managed conservatively with physical therapy, activity modifications, a compression sleeve, and over-the-counter analgesics.

# MSK0005 — Structured SOAP Report Comparison

| SOAP Field | Ground Truth | Predicted (Gemini 3.5 Flash) |
|---|---|---|
| **Chief Complaint** | Right elbow pain | Pain on the outside of the right elbow |
| **Symptoms** | pain on the outside of the right elbow \| pain worse with wrist bending \| pain with lifting heavy objects \| aching pain \| sharp pain with mov... | Achy pain on the outside of the right elbow, occasionally sharp with movement \| Exacerbation of pain at the end of the work day \| Pain trigg... |
| **Symptom Onset** | Several months ago | Several months ago, gradually worsening over time |
| **Medical History** | C-section (17 years ago) | Elective C-section 17 years ago |
| **Medications** | Tylenol \| Advil | Ibuprofen \| Tylenol [STT-corrected] \| Advil [STT-corrected] |
| **Allergies** | No known allergies | No known allergies |
| **Exam Findings** | No redness or swelling of the elbow \| No deformity or misalignment \| Pain on palpation of the lateral aspect of the right elbow \| No pain on... | No visible redness, swelling, misalignment, asymmetry, skin changes, or deformities of the right elbow \| Pain elicited on self-palpation of ... |
| **Diagnosis** | Lateral epicondylitis (Tennis elbow) | Lateral epicondylitis (tennis elbow) |
| **Diff. Diagnosis** | — | Elbow fracture |
| **Prescribed Meds** | Continue Tylenol or NSAIDs as needed | — |
| **Treatment Plan** | Activity modifications (avoid repetitive strain/extensor muscle use at work) \| Use of compression sleeve \| Physical therapy \| Apply ice for ... | Activity modifications at work, specifically avoiding repetitive strain on the forearm extensor muscles \| Use of a forearm compression sleev... |
| **Investigations** | — | — |
| **Follow Up** | The condition should resolve in about six months to two years at most, but normally resolves in a couple of months with physical therapy, co... | The condition normally goes away within a couple of months with conservative management (physical therapy, compression, and splints), though... |

# Case Study: RES0050

Model Analysis using Gemini 3.5 Flash

# RES0050 — Speech-to-Text vs Real Transcript

Comparison of Ground Truth dialogue vs Gemini 3.5 Flash STT output (first ~100 words)

## 📄 Real Transcript (Ground Truth)

D: I was wondering if you could tell me what brought you in.
P: Sure, um, I have been coughing a lot. I've been feeling short of breath and I just feel really wheezy, so I thought I should come check it out.
D: OK, when did the cough start?
P: I've always had a cough. Well, I guess I've had a cough for the past five years, but over the past three days it's gotten a lot worse.
D: OK, did anything happen 3 days ago that has brought the cough on?
...

## 🎙️ Speech-to-Text Transcript (Gemini 3.5 Flash)

D: I was wondering if you could tell me what brought you in?
P: Sure. I have been coughing a lot. I've been feeling short of breath, and I just feel really
P: wheezy, so I thought I should come check it out.
D: Okay. When did the cough start?
P: I've always had a cough. Well, I guess I've had a cough for the past five years,
P: but over the past three days, it's gotten a lot worse.
D: Okay. Did anything happen three days ago that has brought the cough on?
...

# RES0050 — Transcript Summary Comparison

Clinical summary extracted from Ground Truth vs Gemini 3.5 Flash STT output (first ~100 words)

## 📄 Ground Truth Summary

A patient with a 5-year history of chronic cough presented for evaluation of worsening cough, increased yellow/green sputum, wheezing, and exertional shortness of breath over the past 3 days. The patient has a 45 pack-year smoking history (quit 2 years ago) and a past medical history of hypertension and diabetes. The differential diagnosis includes viral infection, bacterial infection, COPD exacerbation, and COVID-19, with plans initiated for a COVID-19 PCR nasal swab and chest X-ray.

## 🎙️ Gemini 3.5 Flash Summary (STT)

A patient with a chronic five-year cough presents with a three-day history of acutely worsening cough, yellow-green sputum, wheezing, and exertional dyspnea. Given the patient's age (eligible for pneumococcal vaccine, thus likely over 65) and 45 pack-year smoking history, investigations including a COVID-19 PCR swab and a chest X-ray are planned to differentiate between viral or bacterial infection, COVID-19, and a COPD exacerbation.

# RES0050 — Structured SOAP Report Comparison

| SOAP Field | Ground Truth | Predicted (Gemini 3.5 Flash) |
|---|---|---|
| **Chief Complaint** | Worsening cough, shortness of breath, and wheezing over the past 3 days. | Worsening cough, wheezing, and shortness of breath |
| **Symptoms** | Worsening cough (chronic for 5 years, exacerbated past 3 days, worse in the morning) \| Shortness of breath (worse with walking and day-to-da... | Worsening cough (chronic for 5 years, acutely worsened over the last 3 days, worse in the morning) \| Shortness of breath (exertional, worse ... |
| **Symptom Onset** | 3 days ago for acute exacerbation of symptoms; chronic cough present for 5 years. | 3 days ago for acute worsening of cough and sputum; last night for wheezing; 5 years for chronic cough |
| **Medical History** | Hypertension \| Diabetes | High blood pressure \| Diabetes |
| **Medications** | Beta-blocker \| Metformin \| Calcium supplement \| Vitamin D | Beta blocker \| Metformin \| Calcium \| Vitamin D |
| **Allergies** | — | No known drug allergies |
| **Exam Findings** | — | Patient reports feeling highly wheezy and short of breath \| No clinical physical examination findings documented in transcript |
| **Diagnosis** | — | Acute exacerbation of respiratory symptoms |
| **Diff. Diagnosis** | Viral respiratory infection \| Bacterial respiratory infection \| COPD exacerbation \| COVID-19 infection | Viral respiratory infection \| Bacterial respiratory infection \| COPD exacerbation \| COVID-19 infection |
| **Prescribed Meds** | — | — |
| **Treatment Plan** | Diagnostic workup to determine etiology of acute respiratory exacerbation. | Advised to schedule pneumococcal vaccination with family physician |
| **Investigations** | COVID-19 PCR nasal swab \| Chest X-ray \| Other laboratory/diagnostic investigations | COVID-19 PCR nasal swab \| Chest X-ray |
| **Follow Up** | Set up pneumococcal vaccination with family doctor; follow up for investigation results to get to the bottom of shortness of breath. | Patient to complete COVID-19 PCR swab, chest X-ray, and other planned investigations to identify the primary cause of dyspnea and guide subs... |

# Case Study: GAS0003

Model Analysis using Gemini 3.5 Flash

# GAS0003 — Speech-to-Text vs Real Transcript

Comparison of Ground Truth dialogue vs Gemini 3.5 Flash STT output (first ~100 words)

## 📄 Real Transcript (Ground Truth)

D: What brings you in here today?
P: Yeah hi, I'm I I have a pain in my belly. It started a couple hours ago.
D: OK, and where exactly in your belly do you have this pain?
P: It's mostly up here on the right side.
D: The right side, OK and can you describe what kind of pain is it? Is it like a dull pain or sharp pain, an aching pain?
...

## 🎙️ Speech-to-Text Transcript (Gemini 3.5 Flash)

D: What brings you in here today?
P: Yeah.
P: Hi.
P: I have a pain in my belly.
P: It started a couple of hours ago.
D: Okay.
D: And where exactly in your belly do you have this pain?
P: It's mostly up here on the right side.
D: On the right side?
P: Yeah.
D: Okay.
D: And can you describe what kind of pain is it?
D: Is it like a dull pain, a sharp pain, an aching pain?
P: I guess it's an aching pain, I'd say.
...

# GAS0003 — Transcript Summary Comparison

Clinical summary extracted from Ground Truth vs Gemini 3.5 Flash STT output (first ~100 words)

## 📄 Ground Truth Summary

Patient is presenting with a constant, aching right-sided abdominal pain that has persisted for 6 hours after an initial 2-hour period of intermittent pain. The patient reports a history of similar postprandial discomfort associated with fatty meals and is currently experiencing nausea but no vomiting, fevers, or bowel changes.

## 🎙️ Gemini 3.5 Flash Summary (STT)

A patient presents with acute right upper quadrant abdominal pain that began 8 hours ago, which was initially intermittent but has been constant for the last 6 hours. The patient reports having similar self-resolving pain episodes in the past following fatty meals and describes currently associated mild nausea, fatigue, and decreased appetite. The medical history is significant for diabetes mellitus controlled with metformin, and the patient's father has a history of a heart attack in his 60s.

# GAS0003 — Structured SOAP Report Comparison

| SOAP Field | Ground Truth | Predicted (Gemini 3.5 Flash) |
|---|---|---|
| **Chief Complaint** | right-sided abdominal pain | Right-sided abdominal pain |
| **Symptoms** | aching pain in right side of abdomen \| nausea \| fatigue \| decreased appetite | aching abdominal pain \| nausea \| fatigue \| decreased appetite |
| **Symptom Onset** | 8 hours ago | Started 8 hours ago; initially came and went for the first 2 hours, but has been constant for the past 6 hours. |
| **Medical History** | diabetes | Diabetes \| Wisdom teeth extraction as a teenager \| Family history of paternal heart attack in his 60s |
| **Medications** | metformin | Metformin |
| **Allergies** | — | No known drug allergies \| No environmental allergies |
| **Exam Findings** | — | Pain localized to the right upper side of the abdomen (8/10 severity) |
| **Diagnosis** | — | Abdominal pain, right upper quadrant |
| **Diff. Diagnosis** | indigestion | Indigestion \| Biliary colic \| Cholecystitis |
| **Prescribed Meds** | — | — |
| **Treatment Plan** | — | Further diagnostic evaluation of right upper quadrant abdominal pain |
| **Investigations** | — | — |
| **Follow Up** | — | — |

# Case Study: DER0001

Model Analysis using Gemini 3.5 Flash

# DER0001 — Speech-to-Text vs Real Transcript

Comparison of Ground Truth dialogue vs Gemini 3.5 Flash STT output (first ~100 words)

## 📄 Real Transcript (Ground Truth)

P: Hey Doc, I'm a 45 year old man coming in, I'm worried there's a little old rash on my leg that I'm getting nervous about.
D: Hi, so you I understand you have a rash on your leg.
P: I don't really know what I would call it. That's what I told the nurse when I came in, but it more hurts than anything.
D: OK, uh, how long is this been going on for?
...

## 🎙️ Speech-to-Text Transcript (Gemini 3.5 Flash)

P: Hey doc, I'm a 45 year old man coming in. I'm worried there's a little rash on my leg I'm getting nervous about.
D: Hi. So I understand you have a rash on your leg.
P: I don't really know what I would call it. That's what I told the nurse when I came in, but it's more hurts than anything.
D: Okay. How long has this been going on for?
P: It's been, it's been looking weird for about a week, but only, yes, it started hurting yesterday.
...

# DER0001 — Transcript Summary Comparison

Clinical summary extracted from Ground Truth vs Gemini 3.5 Flash STT output (first ~100 words)

## 📄 Ground Truth Summary

A 45-year-old male with poorly controlled diabetes presents with a painful, swollen, and red rash on his right ankle of one week's duration, now with purulent drainage. Patient also reports constitutional symptoms including fatigue, increased hunger, and a subjective feeling of fever over the last 12 hours.

## 🎙️ Gemini 3.5 Flash Summary (STT)

A 45-year-old male with poorly controlled Type 2 Diabetes Mellitus presents with a right ankle lesion that became painful, swollen, and increasingly erythematous yesterday, accompanied by a subjective fever and white drainage from an open wound. He reports polyuria, polyphagia, fatigue, and baseline peripheral neuropathy with a suspected active diabetic ulcer on his left heel. A detailed physical examination of his lower extremities will be conducted, followed by clinical consultation with the supervising physician to evaluate for lower extremity cellulitis or a deep wound infection.

# DER0001 — Structured SOAP Report Comparison

| SOAP Field | Ground Truth | Predicted (Gemini 3.5 Flash) |
|---|---|---|
| **Chief Complaint** | Painful, strange-looking rash on right ankle. | Painful, swollen, and red lesion on the right ankle |
| **Symptoms** | rash on right ankle \| pain \| swelling \| erythema \| scabbing \| purulent drainage \| subjective fever/feeling hot \| fatigue \| increased hunger ... | Right ankle pain \| Right ankle redness (erythema) \| Right ankle swelling \| White discharge (leaking) from open ankle wound \| Subjective feve... |
| **Symptom Onset** | Rash began 1 week ago, worsening/painful since yesterday, purulent drainage started within last 12 hours. | The right ankle lesion has appeared abnormal for approximately one week, with pain, swelling, and spreading redness starting yesterday, and ... |
| **Medical History** | Diabetes mellitus (poorly controlled, A1C ~9) \| Hypercholesterolemia \| History of asthma \| Chronic back pain \| History of sinusitis/sinus su... | Type 2 Diabetes Mellitus (uncontrolled, last HbA1c ~9.x) \| Hypercholesterolemia (high cholesterol) \| Childhood asthma \| Chronic back pain (f... |
| **Medications** | Rosuvastatin \| Metformin \| Two other unspecified diabetes medications | Rosuvastatin [STT-corrected] \| Metformin \| Two other unnamed oral diabetes medications \| Tylenol [Acetaminophen] (as needed for back pain) \|... |
| **Allergies** | Penicillin (reaction: facial bloating, respiratory distress) | Penicillin [STT-corrected] (Reaction: facial swelling/bloat and difficulty breathing in childhood) |
| **Exam Findings** | Right ankle appears swollen and erythematous \| Open wound/leaking of purulent material present \| Right leg feels warmer to the touch compare... | Right ankle: swollen, erythematous with spreading redness, and subjectively warm to the touch (hotter than the left leg) \| Right ankle: open... |
| **Diagnosis** | — | Right lower extremity cellulitis, suspect secondary to a diabetic wound infection \| Type 2 Diabetes Mellitus, poorly controlled \| Diabetic n... |
| **Diff. Diagnosis** | Infected diabetic ulcer \| Cellulitis | Right lower extremity abscess \| Osteomyelitis \| Deep vein thrombosis (DVT) |
| **Prescribed Meds** | — | — |
| **Treatment Plan** | Physical examination of the leg \| Consultation with supervising physician | Perform a comprehensive physical exam of the bilateral lower extremities focusing on the right ankle wound and left heel ulcer \| Discuss cli... |
| **Investigations** | — | — |
| **Follow Up** | Patient to undergo physical examination of the leg immediately. | The clinical team will perform a physical exam of the patient's leg and discuss next steps/treatment options with the supervising physician. |